

Algebra 1
Unit 8
Lesson 1 Homework

Name **Answer Key** _____
Date _____
Period _____

Use the graph of the quadratic equation for questions 1 – 4.

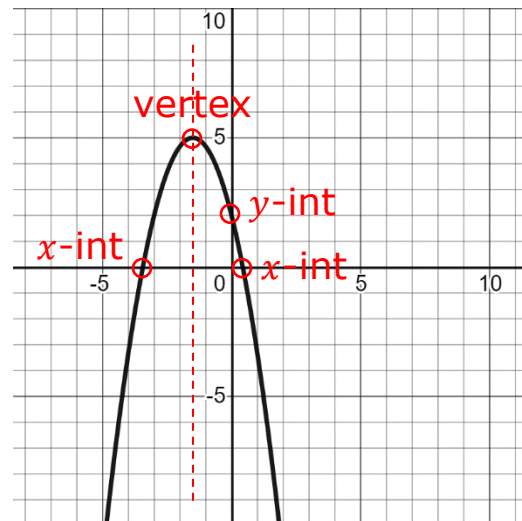
1. Describe where the parabola is increasing.

$(-\infty, -1.5), x < -1.5$

2. Describe where the parabola is decreasing.

$(-1.5, \infty), x > -1.5$

3. Draw the axis of symmetry for the parabola on the graph.



4. Circle the vertex, x -, and y - intercepts of the parabola. Label each point as vertex, x -intercept, and y - intercept.

5. The equations $y = 4(x - 1)(x - 1)$ and $y = 4(x - 1)^2$ both represent the same quadratic. Rewrite both equations in standard form, $y = ax^2 + bx + c$. Show your work.

$y = 4(x - 1)(x - 1)$	$y = 4(x - 1)^2$
$y = (4x - 4)(x - 1)$ $y = 4x^2 - 4x - 4x + 4$ $y = 4x^2 - 8x + 4$	$y = 4(x^2 - 2x + 1)$ $y = 4x^2 - 8x + 4$

6. The equations $y = -(x + 7)(x + 4)$ and $y = -(x - (-7))(x - (-3))$ are equivalent. How can both x -intercepts be determined if the equation $y = -(x + 7)(x + 4)$ is given?

$$x + 7 = 0$$

$$x = -7$$

$$x + 4 = 0$$

$$x = -4$$

Set each factor equal to 0 and solve.

7. The equation $y = \frac{1}{3}x^2 + x - 18$ can be used to determine the y -intercept. Show how each of the *three* forms can be used to find the y -intercept.

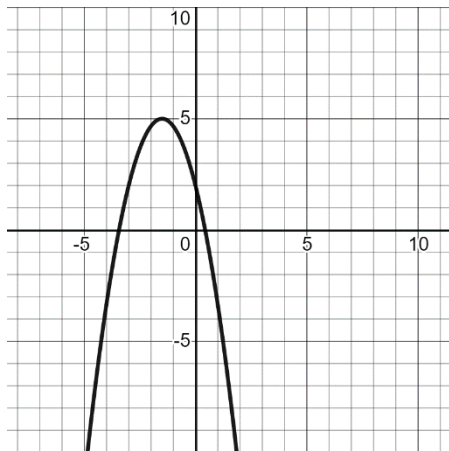
$y = \frac{1}{3}x^2 + x - 18$	$y = \frac{1}{3}(x + 9)(x - 6)$	$y = \frac{1}{3}\left(x + \frac{3}{2}\right)^2 - \frac{225}{12}$
$y = \frac{1}{3}(0)^2 + 0 - 18$ $y = -18$	$y = \frac{1}{3}(0 + 9)(0 - 6)$ $y = \frac{1}{3}(9)(-6)$ $y = -18$	$y = \frac{1}{3}\left(0 + \frac{3}{2}\right)^2 - \frac{225}{12}$ $y = \frac{3}{4} - \frac{225}{12}$ $y = -18$

8. A table of values for $y = -\frac{2}{3}x^2 + \frac{32}{3}x - 42$ is shown. What features can be found using the table of values?

- ☒ x -intercepts
- ☐ y -intercepts
- ☒ vertex
- ☒ axis of symmetry
- ☒ increasing/decreasing
- ☒ positive/negative

x	y
4	-10
5	-5.333
6	-2
7	0
8	0.667
9	0
10	-2

9. The graph of a quadratic equation is shown. Use words, ordered pairs, or symbols to describe the location of each key feature.



Key Feature	Location of Key Feature
x -intercepts	$(-3.5, 0)$ $(0.5, 0)$
y -intercepts	$(0, 2)$
vertex	$(-1.5, 0.5)$
axis of symmetry	$x = -1.5$
increasing	$(-\infty, -1.5), x < -1.5$
decreasing	$(-1.5, \infty), x > -1.5$

10. A table of values for a quadratic equation shown. What features can be found using the table of values?

- ☐ x -intercepts
- ☒ y -intercepts
- ☐ vertex
- ☐ axis of symmetry
- ☐ increasing/decreasing
- ☐ positive/negative

x	y
-1	16
0	15
1	12
2	7

Algebra 1
Unit 8
Lesson 2 Homework

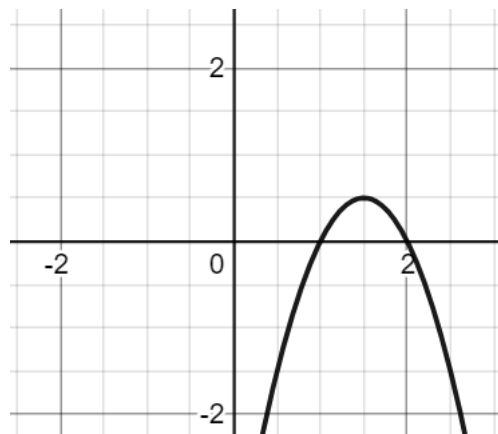
Name **Answer Key** _____
Date _____
Period _____

1. As the x - values of the parabola get infinitely *larger*, the y -values of the parabola approach

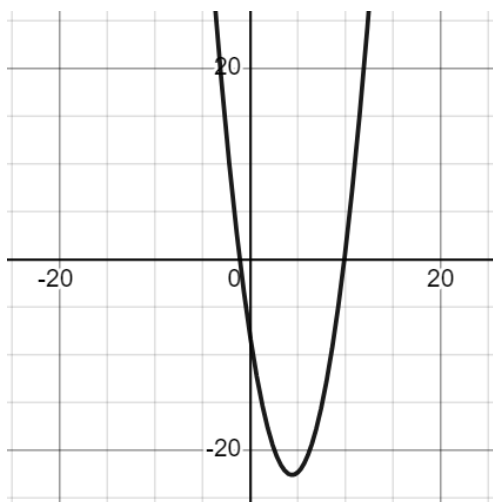
☐ ∞
☒ $-\infty$

2. As the x -values of the parabola get infinitely *smaller*, the y -values of the parabola approach

☐ ∞
☒ $-\infty$



Use the graph of the quadratic equation for questions 3 and 4.



3. Use words or symbols to describe the domain of the quadratic function.

All real #'s

4. Use words or symbols to describe the range of the quadratic function.

$y \geq -22$

5. Determine the vertex and y -intercept of a quadratic using the equation $y = -5(x - 1)^2 + 3$

$y = -5(0 - 1)^2 + 3$
 $y = -2$

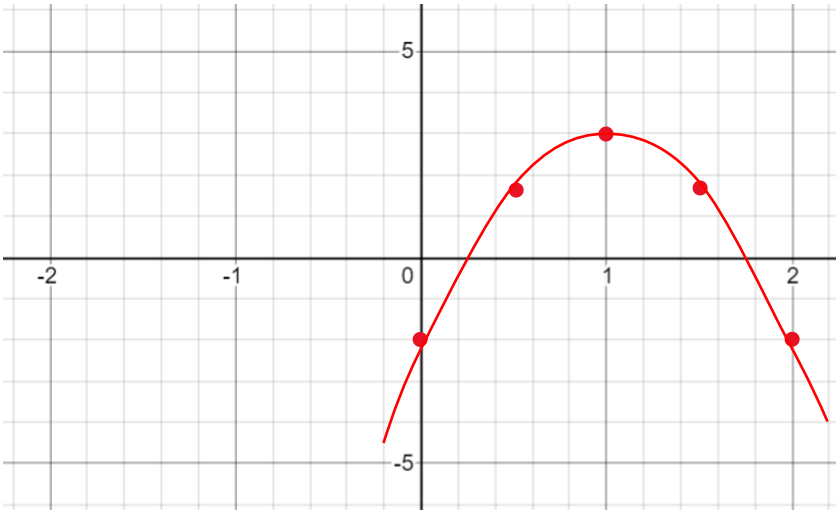
vertex = (1,3)
 y -int = (0, -2)

6. Use the table of values to determine the vertex of the parabola.

vertex = (1,3)

x	y
0.1	-1.05
0.2254	0
0.5	1.75
1	3
1.5	1.75
1.7746	0
2	-2

7. Use the key features you determined in question 5 and 6 to sketch the graph of the quadratic function.



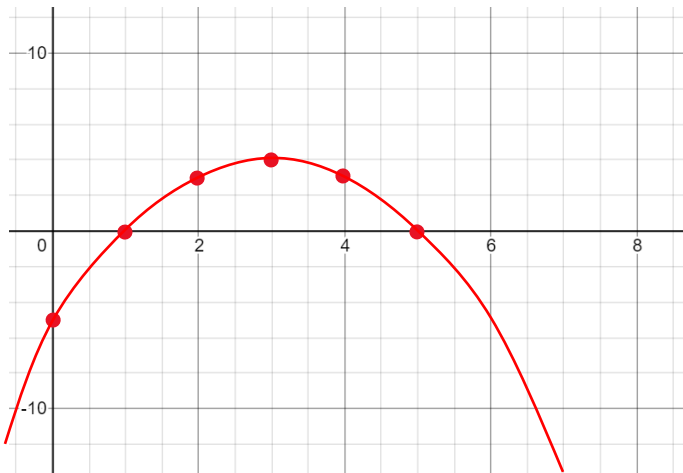
8. Identify key features of the quadratic $y = -(x - 3)^2 + 4$

x -intercept	(1,0)	vertex	(3,4)	domain	All Real #’s	left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{-\infty}$
x -intercept	(5,0)	increasing	$(-\infty, 3)$	range	$y \leq 4$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{-\infty}$
y -intercept	(0,−5)	decreasing	$(3, -\infty)$	opens	Down		

9. Create a table of values for the quadratic function in question 8.

x	−2	−1	0	1	2	3
y	−21	−12	−5	0	3	4

10. Use the key features and table of values from 8 and 9 to graph the quadratic.



Complete the table for each equation in 1-5.

Equation	Form	Key Feature(s) Revealed	Value(s) of Key Feature
1. $y = (x - 7)(x + 5)$	☐ Vertex form ☒ Factored form ☐ Standard form	x -intercepts Opening	(7,0) (-5,0) Up
2. $y = (x - 11)^2 - 23$	☒ Vertex form ☐ Factored form ☐ Standard form	Vertex Axis of symmetry Decrease Increase Opening	(11, -23) $x = 11$ (-∞, 11) (11, ∞) Up
3. $y = -\frac{3}{8}x^2 - \frac{1}{4}x - 2$	☐ Vertex form ☐ Factored form ☒ Standard form	y -intercept Opening	(0, -2) Down
4. $y = -13(x + 12)^2$	☒ Vertex form ☐ Factored form ☐ Standard form	Vertex Opening Axis of symmetry Increase Decrease	(-12, 0) Down $x = -12$ (-∞, -12) (-12, -∞)
5. $y = -x^2 + 11x - 52$	☐ Vertex form ☐ Factored form ☒ Standard form	y -int Opening	(0, -52) Down

6. Rewrite the quadratic equation in factored form.

$$y = x^2 + x - 56$$

$$y = (x + 8)(x - 7)$$

7. Rewrite the quadratic equation in standard form.

$$\begin{aligned}y &= (x - 9)^2 - 21 \\y &= x^2 - 18x + 81 - 21 \\y &= x^2 - 18x + 60\end{aligned}$$

8. Rewrite the equation in factored form.

$$\begin{aligned}y &= x^2 + 6x - 91 \\y &= (x + 13)(x - 7)\end{aligned}$$

9. Rewrite the equation in standard form.

$$\begin{aligned}y &= -7(x - 6)(x + 8) \\y &= -7(x^2 + 2x - 48) \\y &= -7x^2 - 14x + 336\end{aligned}$$

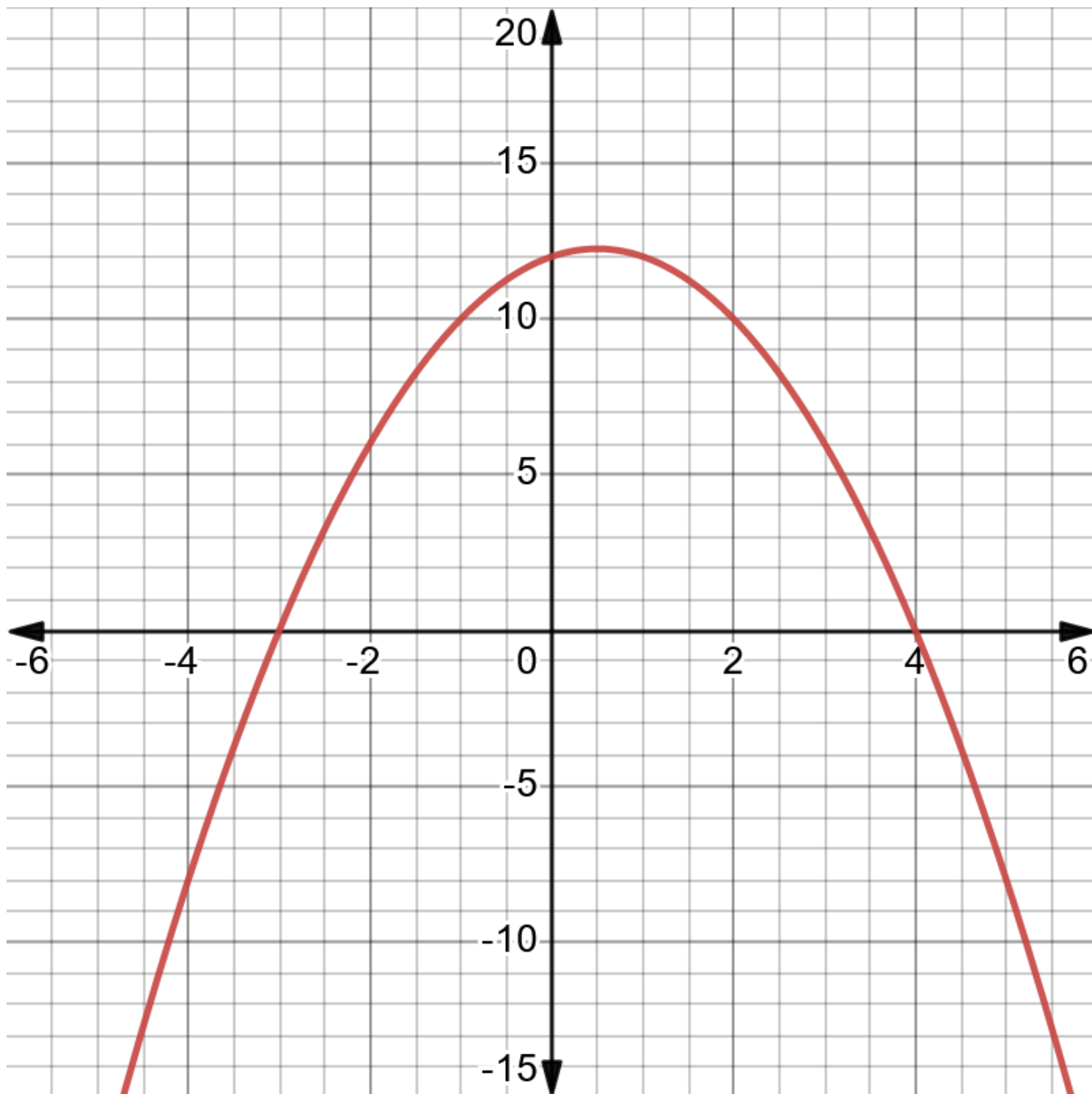
10. Laila is playing with a slingshot and shoots a ball straight up in the air. The ball is shot straight up from 2 meters above the ground with a velocity of 4 meters/second. The height of the ball is represented by the equation $y = -6x^2 + 4x + 2$, where x is time in seconds. Rewrite the equation in factored form.

$$\begin{aligned}y &= -2(3x^2 - 2x - 1) \\y &= -2(3x + 1)(x - 1)\end{aligned}$$

Algebra 1
Unit 8
Lesson 4 Homework

Name **Answer Key** _____
Date _____
Period _____

1. Graph the equation $y = -(x - 4)(x + 3)$.



2. Write the key features that can be determined from the equation.

$y = -(x - 4)(x + 3)$	
x -intercept	$(-3,0) (4,0)$
y -intercept	$(0,12)$
Vertex	$(0.5,12.25)$
Domain	All real #'s
Range	$y \leq 12.25$
Left end behavior	$x \rightarrow -\infty \quad y \rightarrow -\infty$
Increasing	$x < 0.5$
Decreasing	$x > 0.5$
Positive	$-3 < x < 4$
Negative	$x < -3, x > 4$
Opens up/down	Down
Axis of symmetry	$x = 0.5$
Right end behavior	$x \rightarrow \infty \quad y \rightarrow -\infty$

Re-write each expression into vertex form by completing the square.

$$\begin{aligned} 1. \quad & x^2 + 14x - 38 \\ & [x^2 + 14x + 49] - 38 - 49 \\ & (x + 7)(x + 7) - 87 \\ & (x + 7)^2 - 87 \end{aligned}$$

$$\begin{aligned} 2. \quad & n^2 - 2n - 3 \\ & [n^2 - 2n + 1] - 3 - 1 \\ & (n - 1)^2 - 4 \end{aligned}$$

$$\begin{aligned} 3. \quad & f^2 + 14f - 51 \\ & [f^2 + 14f + 49] - 51 - 49 \\ & (f + 7)^2 - 100 \end{aligned}$$

$$\begin{aligned} 4. \quad & a^2 + 2a + 20 \\ & [a^2 + 2a + 1] + 20 - 1 \\ & (a + 1)^2 + 19 \end{aligned}$$

$$\begin{aligned} 5. \quad & b^2 - 4b + 4 \\ & b^2 - 4b + 4 \\ & (b - 2)^2 \end{aligned}$$

$$\begin{aligned} 6. \quad & c^2 + 6c + 8 \\ & [c^2 + 6c + 9] + 8 - 9 \\ & (c + 3)^2 - 1 \end{aligned}$$

$$7. \quad \begin{array}{l} g^2 + 4g + 4 \\ (g + 2)^2 \end{array}$$

$$8. \quad \begin{array}{l} t^2 - 10t \\ [t^2 - 10t + 25] - 25 \\ (t - 5)^2 - 25 \end{array}$$

$$9. \quad \begin{array}{l} z^2 + 4z - 6 \\ [z^2 + 4z + 4] - 6 - 4 \\ (z + 2)^2 - 10 \end{array}$$

$$10. \quad \begin{array}{l} d^2 + 6d + 8 \\ [d^2 + 6d + 9] + 8 - 9 \\ (d + 3)^2 - 1 \end{array}$$

Re-write each expression into vertex form by completing the square.

1. $5x^2 - 20x + 6$

$$5(x^2 + 4x + 4) + 6 - 20$$

$$5(x - 2)^2 - 14$$

2. $3x^2 - 6x - 34$

$$3(x^2 - 2x + 1) - 34 - 3$$

$$3(x - 1)^2 - 37$$

3. $3y^2 - 6y - 3$

$$3(y^2 - 2y + 1) - 3 - 3$$

$$3(y - 1)^2 - 6$$

4. $2b^2 + 8b - 7$

$$2(b^2 + 4b + 4) - 7 - 8$$

$$2(b + 2)^2 - 15$$

5. $4p^2 + 16p - 15$

$$4(p^2 + 4p + 4) - 15 - 16$$

$$4(p + 2)^2 - 31$$

6. $9a^2 - 18a - 21$

$$9(a^2 - 2a + 1) - 21 - 9$$

$$9(a - 1)^2 - 30$$

$$\begin{array}{l}
 7. \quad 3r^2 - 12r + 7 \\
 \quad 3(r^2 - 4r + 4) + 7 - 12 \\
 \quad 3(r - 2)^2 - 5
 \end{array}$$

$$\begin{array}{l}
 8. \quad 3z^2 - 18z + 5 \\
 \quad 3(z^2 - 6z + 9) + 5 - 27 \\
 \quad 3(z - 3)^2 - 22
 \end{array}$$

$$\begin{array}{l}
 9. \quad 2t^2 - 12t - 1 \\
 \quad 2(t^2 - 6t + 9) - 1 - 18 \\
 \quad 2(t - 3)^2 - 19
 \end{array}$$

$$\begin{array}{l}
 10. \quad 3f^2 - 12f - 9 \\
 \quad 3(f^2 - 4f + 4) - 9 - 12 \\
 \quad 3(f - 2)^2 - 21
 \end{array}$$

Algebra 1

Unit 8

Lesson 7 Homework

Name **Answer Key** _____

Date _____

Period _____

For the expression, the first step to completing the square is shown.

$$\begin{aligned} x^2 - 8x + 15 \\ (x^2 - 8x) + 15 \end{aligned}$$

1. Complete the perfect square trinomial.

$$(x^2 - 8x + 16) + 15 - 16$$

$$(x - 4)^2 - 1$$

2. In the expression, the value that was added to create the perfect square trinomial is being multiplied by 1, so the value that should be subtracted from 15 to maintain equivalency is 16.

Complete the table to complete the square without using algebra tiles.

	Process	Equation
	Original equation	$y = -8x^2 - 16x + 42$
	Factor out the common factor of the first two terms.	$y = -8(x^2 + \underline{2}x) + 42$
3.	Determine the value that makes a perfect square trinomial in the parentheses.	$y = (x^2 + \underline{2}x + \underline{1}) + 42$
4.	Then, to maintain equivalence, multiply the factor by the value. Add the opposite of that value.	$y = -8(x^2 + \underline{2}x + \underline{1}) + 42$ $\quad \quad \quad + \underline{8}$
5.	Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = -8(x + \underline{1})^2 + 42 + \underline{8}$
6.	Combine like terms.	$y = -8(x + \underline{1})^2 + \underline{50}$

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

7. $y = x^2 - 10x - 24$

$y = (x - 5)^2 - 49$

$y = (x - 12)(x + 2)$

8. Key Features

Key Features					
x-intercept	(12,0)	vertex	(5, -49)	domain	All Real
x-intercept	(-2,0)	increasing	$x > 5$	range	$y \geq -49$
y-intercept	(0, -24)	decreasing	$x < 5$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\infty}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\infty}$			

9. $y = 6x^2 + 12x - 48$

$y = 6(x^2 + 2x + 1) - 48 - 6$

$y = 6(x + 1)^2 - 54$

10. Key Features

Key Features					
x-intercept	(-4,0)	vertex	(-1, -54)	domain	All Real
x-intercept	(2,0)	increasing	$x < -1$	range	$y \geq -54$
y-intercept	(0, -48)	decreasing	$x > -1$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\infty}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\infty}$			

Algebra 1
Unit 8
Lesson 8 Homework

Name **Answer Key** _____
Date _____
Period _____

In 1965, a Ford Mustang cost \$2,600. While most cars lose value, a 1965 Mustang in immaculate condition can be worth over \$30,000. The function $M(x) = 20x^2 - 420x + 2600$ models the price of the 1965 Mustang in the United States since 1965.

1. What does the variable x represent? Include the units.

Years since 1965

2. What does the variable y represent? Include the units.

Value of the Mustang, in dollars

3. Interpret the y -intercept. Include units.

(0,2600) In 1965, the Mustang was worth \$2600

4. Rewrite the function in vertex form. Show your work.

$$M(x) = 20(x^2 - 21x + 110.25) + 2600 - 2205$$

$$M(x) = 20(x - 10.5)^2 + 395$$

5. Interpret the vertex in terms of the context. Include units.

(10.5,395)

The Mustang's lowest value was \$395, 10.5 years after 1965.

6. Complete the table of values for $M(x)$.

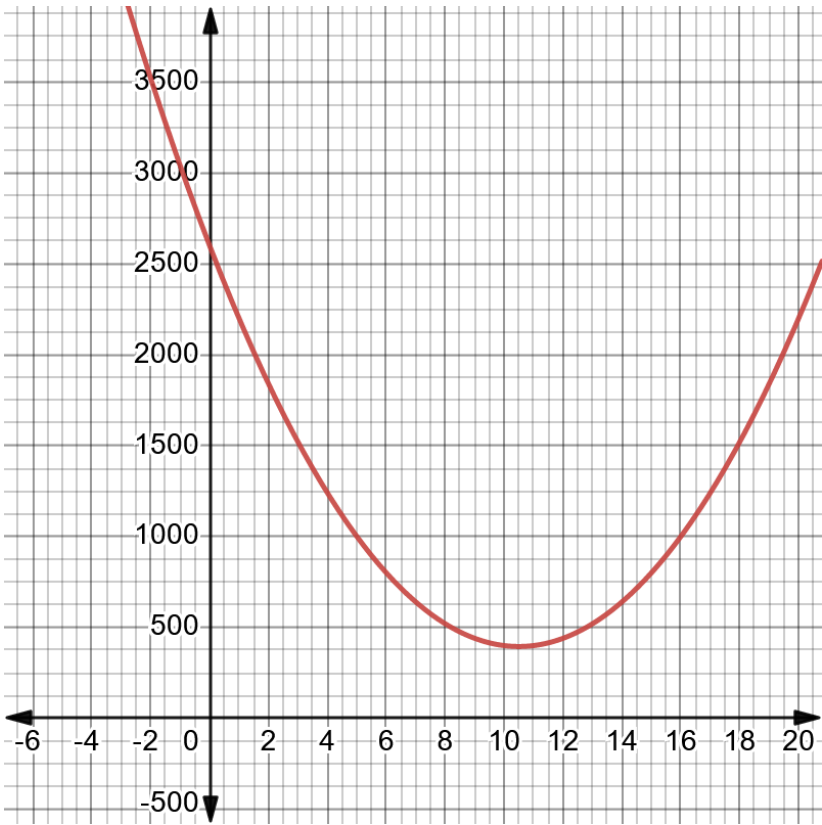
x	-5	-3	0	7	10.5	12	24	50
$M(x)$	5200	4040	2600	640	395	440	4040	31600

7. Circle the values in the table that do not make sense in terms of the context.

8. Complete as many of the key features as possible using the given function, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x-intercept		vertex	(10.5,395)	domain	All Real
x-intercept		increasing	$x > 10.5$	range	$y \geq 395$
y-intercept	(0,2600)	decreasing	$x < 10.5$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \infty$			
right end behavior		$x \rightarrow \infty, y \rightarrow \infty$			

9. Graph the function.



10. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set builder notation.

$$\{x|0 \leq x \leq 60\} \quad \{y|y \geq 395\}$$

Algebra 1
Unit 8
Lesson 9 Homework

Name **Answer Key**
Date _____
Period _____

A theater owner is looking to maximize his profits and needs to increase his ticket prices. The function $P(t) = -50t^2 + 100t + 14,400$ represents his total profit, P , based on the amount, in dollars, he raises the ticket price, t .

1. Rewrite the function in vertex form.

$$P(t) = -50(t - 1)^2 + 14,450$$

2. Rewrite the function in factored form.

$$P(t) = -50(t - 18)(t + 16)$$

Complete the statements.

3. The

☐ vertex
☒ **factored**
☐ standard

 form of the function can be used to determine the interval on which the theater owner experiences a profit. The theater owner experiences a profit on the interval $0 \leq t \leq 18$.

4. The

☐ **vertex**
☐ factored
☐ standard

 form of the function can be used to determine when the theater owner experiences the greatest profit. The theater owner experiences the greatest profit when $t = 1$.

5. The

☐ vertex
☒ **factored**
☐ standard

 form of the function can be used to determine when the theater owner experiences neither a profit nor a loss. This occurs when the ticket price increased by \$18.

The function $H(x) = 0.5x^2 - x + 33$ models the amount of people, in millions, H , enrolled in an HMO insurance in the United States in the years, x , since 1990.

6. What would the zeros of this function reveal in this context?

The number of years since 1990 when there would be 0 people enrolled in the plan.
Sincer there are no zeros, there are no years with 0 people enrolled.

7. Rewrite the function in vertex form.

$$H(x) = 0.5(x - 1)^2 + 32.5$$

Complete the statements.

8. The

☒ vertex
☐ factored
☐ standard

 form of the function can be used to determine the year with the lowest HMO enrollment.

9. The year with the lowest HMO enrollment was 1991 when 32.5 million people were enrolled.

10. The

☐ vertex
☐ factored
☒ standard

 form of the function can be used to determine how many people, in millions, were enrolled in an HMO in 1990, 33 million people.